DATA 582 Project Part 4: Prototype & Evaluation

This report presents the first protoyype for the interactive dashboard and a user evaluation study for the Calgary Crime and Population analysis project, focusing on design decisions, usability, and interpretation of crime patterns across communities.

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Project: Exploring Crime Patterns in Calgary (2018-2024): An Interactive Dashboard Evaluation

Due Date: April 2nd, 2026 (24 hour extension given)

Dashboard Link: https://muntahaltaie.github.io/DATA-582-PROJECT-/

Repository Link: https://github.com/muntahaltaie/DATA-582-PROJECT-

Dataset Description (From Part 2)

For this project, I chose to extract data from the City of Calgary Open Data Portal. My topic analyzes crime and population data in Calgary and how reported crime varies across communities. My project proposal aimed to explore how crime is distributed across different neighbourhoods and how population dynamics may shape the trends we observe. The primary dataset used was the City of Calgary Community Crime Statistics. This dataset reports aggregated monthly crime counts by attributes like community, year, and crime category. I selected this dataset because it provides a comprehensive look at multi-year crime data at the community level versus citywide totals. This makes it possible to analyze the data based on spatial differences and temporal trends.

I also used the City of Calgary Census by Community 2019 dataset. This dataset includes attributes like population, quadrant, and community structure information for each neighbourhood. I chose this dataset to add

additional context to the density observed within neighborhoods. For example, smaller neighbourhoods with 100 crimes should be interpreted differently from a large neighbourhood with 500 crimes. Although the census is from 2019, it is the most recent complete community-level population dataset available from the city. For this reason, a population baseline was established on the 2019 census data.

I originally was planning on incorporating a third dataset: The Historical Community Populations dataset was going to be used in order to provide context for the crime data by showing how many people live in each community and how this changes over time. However, since it did not have complete data for all communities and years in my crime dataset, using it would have created gaps and required making assumptions about missing population values. Instead, I chose to use the 2019 Census by Community dataset to create a population baseline so I could calculate crime rates with population data while limiting the amount of uncertainty in this project. By combining these two datasets, I created a dataset that analyzes how much crime is reported in each community as well as how crime relates to the number of people living in these communities. This supports my original project goals of comparing crime patterns across communities with different population sizes and examining how reported crime changes over time in a growing city like Calgary.

Dashboard Description

The project iteration features the first prototype for a interactive dashboard that explores crime patterns across Calgary from 2018 to 2024. It was created in order to help users compare crime counts and crime rates per 1,000 residents across communities, examine how crime changes over time, and analyze how crime rates vary across different sectors of the city. The dashboard brings together the main design ideas developed in earlier project iterations into one linked and interactive view.

The dashboard includes three main visual components. The first is a bar chart that highlights the top communities based on either total crime count or crime rate. This allows users to compare how rankings change when population is taken into account. The second is a time series chart that shows how crime levels change

over time under the current filter settings. The third is a boxplot that summarizes the distribution of crime rates across Calgary sectors. This helps users compare sector variation and spread rather than only average values.

Users can interact with the visualization through filters for crime type, sector, and year, as well as a metric toggle that switches between raw counts and normalized crime rates. These interactions were included so users can draw comparisans beyond static visual summaries while exploring how crime patterns change under different conditions.

Dashboard Features

The dashboard includes interactive filters for crime type, sector, year, and community, so users can focus on specific parts of the data for analysis. I also added a searchable community filter to make it easier to find individual neighbourhoods. There is also a metric toggle that lets users switch between crime count and crime rate per 1,000. This helps compare both total crime and population adjusted crime levels. The bar chart shows the top communities under selected filters. Users can click a bar to focus on that community in the trend chart. The line chart shows how crime changes over time. Lastly, the boxplot compares how crime rates are distributed across different sectors across the City of Calgary. Together, these features make the dashboard more interactive and allow users to explore/interact with the data from multiple perspectives.

Intended Audience

The intended audience for this dashboard is a general audience with basic visualization knowledge including students, Calgary residents, and anyone interested in understanding crime patterns across the city. The visualization was created for those less familiar with data science topics/visuals. It aims to make community level crime data more accessible by using intuitive chart types, interactive filters, and direct comparisons between crime counts and crime rate metrics. Because of this audience, the design and evaluation for the initial

prototype of the dashboard will focus on readability, basic interactions, and helping users explore the data without the need for high level statistical knowledge.

Questions

The main goal of this evaluation was to understand whether the dashboard effectively communicates crime patterns across Calgary and whether users can correctly interpret the relationship between crime counts and population. The following questions guided the study evaluation:

1. Can users accurately distinguish between total crime counts and crime rates when using the dashboard?: This is important because one of the main design challenges in the project was avoiding misleading interpretations caused by population differences across communities.

2. Can users easily navigate the dashboard and use its filters to explore different crime trends?: Since the final prototype includes interactive filtering by community, crime type, and year, it is important to evaluate whether users can actually use these features without confusion.

3. Are users able to extract meaningful insights about crime patterns, such as identifying high risk areas or changes over time?: This helps determine whether the visualization is not only usable but also actually useful.

In order to answer these questions I used a couple of evaluation methods. To answer these questions, I used a combination of three evaluation methods.

1. Observation: I watched participants complete tasks and noted where they hesitated or made errors.

2. Measurement: I recorded whether tasks were completed correctly and approximately how long they took.

3. Post-task questionnaire and short interview: Allowed participants to explain their understanding and provide feedback on clarity and usability.

Method

The evaluation was conducted on a small group of participants who had some basic understanding of graphs but no experience with this specific dataset. This matches my intended audience for the visualizations, which is

general users rather than those well-versed in data science visuals. Participants were recruited informally through classmates and peers. All participation was voluntary. Each session lasted roughly 20-30 minutes and followed a uniform structure to limit variability within the results. At the beginning of the evaluation, participants were given a brief explanation of the purpose of the dashboard and how it relates to crime patterns in Calgary. Following this, they were then asked to complete a set of predefined tasks within the dashboard that remained uniform through each evaluation. Some of the tasks were designed to test the research questions listed above. These tasks included the following:

- Identifying the community with the highest total crime count
- Switching from crime count to crime rate (per 1,000 residents) and identifying which communities had the highest rates
- Temporal Trend analysis and comparison on how crime trends change between 2018 and 2024
- Using the filters within the dashboard (e.g., crime type, sector, year) to identify and analyze specific patterns

During these tasks, participants were encouraged to think out loud and explain what they were doing and how they were interpreting the charts. I observed these interactions and made notes on any errors/incorrect responses, hesitation/confusion, usability of filters and controls.

For the measurement portion of the evaluation, I recorded whether each task was completed correctly, the approximate time taken per task, and the number of attempts/corrections.

Lastly, I had the participants fill out a short questionnaire that included questions like:

- “Was the difference between crime count and crime rate clear?”
- “Which chart was easiest to understand?”

- “Which part of the dashboard was confusing?”

Risks

The study involved minimal risk since the participants were interacting with a digital dashboard and answering questions about their experience. However, there were still some potential risks that should be addressed

One risk is that participants may feel uncomfortable during the evaluation while completing the tasks, knowing that they were being recorded. To address this, participants were clearly informed that the purpose of the study was to evaluate the usability of the dashboard and not their performance. Additionally, they were encouraged to take their time and were reminded that there are "no wrong answers."

Another risk was the privacy of the participants. No personal identifiers were collected, and participants were not asked to share any sensitive data. All responses were anonymized and labeled as "Participant x."

There was no pressure to participate in the study; participants were recruited casually through acquaintances. Consent for evaluation was obtained using a consent form. Participants were informed that they could withdraw at any time with no consequences. While screen recordings and audio were collected, no recordings of faces or any personal identifying features were taken. Notes regarding the evaluation were stored in a secure document requiring two-factor authentication by the student researcher and were used only for the purpose of this assignment.

Results

The results show that there were patterns across both task performance and participant feedback. For tasks surrounding total crime counts, the participants were able to correctly identify the highest ranking communities with little to no difficulty. These tasks were completed quickly with accurate responses. This suggests that the

bar charts representing the crime counts were intuitive to the participants.
Performance dropped when participants were asked to interpret the crime rates versus the crime counts. Only about 60% of participants were able to correctly identify the highest crime rate community on their first attempt at the task. The others initially selected communities with high counts rather than high rates. This indicates that there may be some confusion between the two metrics, which highlights the challenges of communicating normalized data to the participating audience.

When looking at the usability of the dashboard, all participants were able to locate the filters and metric toggle, but some took longer than others. Several participants initially interacted with the charts without applying a filter, and later on realized that it impacted the results. This may suggest that the interaction design is functional but not as intuitive.

The boxplot visuals had a mix of results. Some participants were able to identify the variability across sectors, but most struggled to interpret what the boxplot represented. This suggests that more complex statistical visuals may require additional explanation to the audience.

From the questionnaire, participants generally described the dashboard as visually clean and informative, but noted that:

- The difference between count and rate needs more explanation
- Filter interactions were not immediately obvious
- The boxplot requires more guidance

Disscusion

The evaluation results gave me a clear direction on how to improve the final version of the dashboard in Project Part 5. One of the most important findings yielded from the evaluation was that users struggled to distinguish

between total crime counts and crime rates, even though both of these measures were included in the design. While the metric toggle and side-by-side comparisons did help, I found that most participants automatically defaulted to interpreting the raw counts, which indicates a usability issue. To address this going forward, I plan to make the distinction between the metric more explicit by adding short explanatory tooltips as well as annotations directly onto the visuals. An example of this could be including a small note that explains why the crime rate is often a more purposeful measure versus total counts, especially in cases when we are comparing communities of different sizes.

Another area I plan to improve on is the clarity surrounding the interactions, specifically with the filters being used. Even though all the participants were eventually able to use the filters, there was initially some hesitation/confusion about how the filters affected the visuals. This suggests that the interaction design is not intuitive to the audience. To fix this, I plan to improve visibility and feedback of the filters by adding clearer labels, highlighting which filters are active, and potentially including a short explanation at the top of the dashboard. Adding animations/slow transitions when filters are applied can also help show users how the data is changing. Lastly, an option to display graphs side by side as well as individually may also improve the intuitiveness of the filtering system.

The boxplot visualization was the least intuitive component for the participants in the dashboard. Boxplots are good as they provide us with statistical information about variability across sectors. However, with the intended audience used for this study, it may be difficult to interpret without basic statistical knowledge. To improve this, I plan to either simplify the visual or support it with additional features like including a short description of what the box plot represents, labeling key statistics like the median and the range, or including a notation that highlights the key takeaway from the visual (e.g., which sector has the highest variability)

In addition to these changes, I also plan to improve insight visibility by adding direct annotations to the charts. It's important to note that during the evaluation, some participants were able to identify some trends, but missed other key patterns. By highlighting these important insights (e.g., peak crime years, sectors with high variability, etc.) I can help direct the user's analysis without introducing bias or the exploratory aspect of the dashboard.

This evaluation confirmed that the dashboard is functional and visually clear, but will require improvements to clarity, guidance, and interaction design. The final iteration of this project will focus on making the dashboard more intuitive for first-time users while strengthening and reinforcing its ability to communicate meaningful insights and trends observed from the data.

Resources

I used the City of Calgary Open Data Portal datasets as the main data sources for this project, specifically the Community Crime Statistics dataset and the Census by Community 2019 dataset, while also reviewing the Historical Calgary Community Populations dataset during earlier planning stages. Statistics Canada and Pinkerton were used to support how crime data is commonly reported and why crime rates per capita are often more meaningful than raw counts for comparison. For visualization design, I used to Munzner's Visualization Analysis and Design, Cleveland and McGill's work on graphical perception, and Wickham and Stryjewski's paper on boxplots to support encoding and chart choices. The dashboard itself was built in D3, using the D3 paper and D3 documentation/examples for selections, events, scales, axes, shapes, transitions, the line chart structure, and general chart inspiration. For evaluation design, I used usability resources from Digital.gov, Nielsen Norman Group, and the Interaction Design Foundation to guide task-based testing and participant feedback collection. Ethical procedures for consent, minimal risk, and participant treatment were guided by the TCPS 2 policy and its chapters on the ethics framework and consent process.

GenAI Statement: No generative AI was used for iteration 4 of this project, nor will it be used for any subsequent parts.

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